Inferences Name: _____ Per: _____

In the below, A is an $m \times n$ matrix. For each question, circle the most specific response that can be determined. If no choice can be determined, circle the choice containing "not determined" or "no conclusion".

1) If A represents an onto mapping, and $A\boldsymbol{x} = \boldsymbol{b}$ is considered, that implies

(A) $m \geq n$ (B) there is at least one free variable (C) $A\boldsymbol{x} = \boldsymbol{b}$ is consistent for all $\boldsymbol{b} \in \mathbb{R}^m$

(D) $m > n$ (E) there are no free variables (F) None of these conclusions can be reached

2) Given A, if $A\boldsymbol{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 0 \end{bmatrix}$ has an infinite number of solutions, that implies that $A\boldsymbol{x} = \boldsymbol{b}$

(A) does not have exactly one solution for any $\boldsymbol{b} \in \mathbb{R}^4$ (B) has exactly four components in $\boldsymbol{x}$, if $\boldsymbol{x}$ is a solution (C) has an infinite number solutions for every $\boldsymbol{b} \in \mathbb{R}^4$

(D) has at least one solution for every $\boldsymbol{b} \in \mathbb{R}^4$ (E) No conclusion above can be reached.

3) If A has no free variables, that implies that

(A) $m < n$ (B) $m \leq n$ (C) $m = n$ (D) $m \geq n$ (E) $m > n$ (F) A's size is not determined

4) If $A\boldsymbol{x} = \boldsymbol{b}$ has exactly one solution for each $\boldsymbol{b} \in \mathbb{R}^m$, that implies that the columns of A

(A) are linearly dependent (B) do not all have pivots

(C) do not span $\mathbb{R}^m$ (D) all have pivots (E) are not determined

5) If the augmented matrix $[A \ \ \boldsymbol{b}]$ that corresponds to the system $A\boldsymbol{x} = \boldsymbol{b}$ has a pivot in each column, that implies that

(A) $col(A) = \mathbb{R}^m$ (B) $m < n$ (C) $m \leq n$ (D) $m = n$

(E) $col(A) \neq \mathbb{R}^m$ (F) A is onto (G) None of these conclusions can be reached

6) $A\boldsymbol{x} = \boldsymbol{b}$ has at least one solution for each $\boldsymbol{b} \in \mathbb{R}^m$ implies that the columns of A

(A) are linearly dependent (B) span $\mathbb{R}^m$

(C) are linearly independent (D) do not span $\mathbb{R}^m$ (E) are not determined

7) If A has at least one free variable, that implies that $A\boldsymbol{x} = \boldsymbol{b}$

(A) has at most one solution for each $\boldsymbol{b} \in \mathbb{R}^m$ (B) does not have exactly one solution for any $\boldsymbol{b} \in \mathbb{R}^m$ (C) cannot be solved for some $\boldsymbol{b} \in \mathbb{R}^m$

(D) has at least one solution for each $\boldsymbol{b} \in \mathbb{R}^m$ (E) has infinitely many solutions for each $\boldsymbol{b} \in \mathbb{R}^m$ (F) is not determined

8) If A is given and $A\boldsymbol{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 0 \end{bmatrix}$ has no solutions, that implies that $A\boldsymbol{x} = \boldsymbol{b}$ has

(A) no solution for any $\boldsymbol{b} \in \mathbb{R}^4$ (B) exactly four components in $\boldsymbol{x}$, if $\boldsymbol{x}$ is a solution (C) either no solutions or an infinite number of solutions for each $\boldsymbol{b} \in \mathbb{R}^4$

(D) at most one solution for every $\boldsymbol{b} \in \mathbb{R}^4$ (E) No conclusion above can be reached.

9) If A has a pivot in every column, that implies

(A) $col(A) = \mathbb{R}^m$ (B) $m < n$ (C) $m \leq n$ (D) $m = n$

(E) $col(A) \neq \mathbb{R}^m$ (F) A is 1-to-1 (G) A is onto (H) None of these

10) If $A\boldsymbol{x} = \boldsymbol{b}$ has infinitely many solutions for each $\boldsymbol{b} \in \mathbb{R}^m$, that implies that A has

(A) $m < n$ (B) $m \leq n$ (C) $m = n$ (D) $m \geq n$ (E) $m > n$ (F) A's size is not determined

11) If the columns of A do not span $\mathbb{R}^m$, that implies that A has

(A) a pivot in each row (B) more rows than pivots (C) a pivot in each column

(D) no free variables (E) none of these

12) If A has a pivot in each row, that implies that

(A) $m < n$ (B) $m \leq n$ (C) $m = n$ (D) $m \geq n$ (E) $m > n$ (F) A's size is not determined

13) If the columns of A are linearly independent, that implies that $A\boldsymbol{x} = \boldsymbol{b}$

(A) has at most one solution for each $\boldsymbol{b} \in \mathbb{R}^m$ (B) does not have exactly one solution for any $\boldsymbol{b} \in \mathbb{R}^m$ (C) cannot be solved for some $\boldsymbol{b} \in \mathbb{R}^m$

(D) has at least one solution for each $\boldsymbol{b} \in \mathbb{R}^m$ (E) has infinitely many solutions for each $\boldsymbol{b} \in \mathbb{R}^m$ (F) has exactly one solution for each $\boldsymbol{b} \in \mathbb{R}^m$

(G) We cannot conclude any of the above with certainty

14) $m < n$ implies the columns of A

(A) are linearly dependent (B) span $\mathbb{R}^m$ (C) are linearly independent

(D) do not span $\mathbb{R}^m$ (E) We cannot conclude any of the above with certainty